

Chemistry Course Handbook

Course Code: CHEM-11 | Academic Year 2025–2026

Course Description

Foundational Chemistry covering atomic structure, bonding, reactions, and stoichiometry.

Course Overview

This course introduces the composition and behavior of matter. Laboratory work emphasizes measurement, safety, and quantitative analysis alongside theoretical instruction.

Learning Outcomes

By the end of this course, students will be able to:

- 1 Describe atomic structure and use the periodic table to predict element properties.
- 2 Distinguish between ionic, covalent, and metallic bonding and their resulting properties.
- 3 Balance chemical equations and classify types of chemical reactions.
- 4 Perform stoichiometric calculations involving moles, mass, and gas volumes.
- 5 Apply laboratory safety procedures and accurate measurement techniques.

Assessment

35% labs, 25% tests, 15% assignments, 25% final exam.